Pragya Jha

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EDUCATION

SRM University | *Sonipat, Haryana* — ***2022-2026***

Computer Science and Engineering (with specialization in Data Science and Artificial Intelligence) | ***B.Tech*** — ***CGPA: 7.4***

TECHNICAL SKILLS

- **Languages:** Python, R, JavaScript
- **Frameworks & Technologies:** Django, FastAPI, React, Next.js, REST APIs
- **Data Science & ML:** Machine Learning, Artificial Intelligence (AI), Generative AI, LLM Integration (GPT-4.1 Mini), Pattern Recognition, Data Visualization, Hugging Face, Scikit-learn, Pandas
- **Databases:** SQL, MySQL, PostgreSQL, MongoDB, NoSQL, Azure SQL Database
- **Cloud & DevOps:** Azure (Virtual Machines, Cloud Deployment), IBM Cloud, Git, CI/CD, Postman
- **Core Concepts:** Data Structures & Algorithms, Software Engineering, API Testing, Concurrency Handling, Cloud Application Development, Agile

EXPERIENCE

Freelance - International Client | AI Question Paper Generation System — ***Mar 2026– May 2026***

- Architected and deployed a full-stack automated assessment generation platform using GPT-4.1 Mini, React, and FastAPI, producing syllabus-aligned 50-page question papers in under two clicks for an Australian coaching centre client.
- Designed prompt engineering pipelines supporting 3 distinct test formats, across multiple year levels and difficulty tiers, maintaining 100% structural consistency in generated output.
- Provisioned and managed backend infrastructure on Azure Virtual Machines with Azure SQL Database, resolving concurrency issues and ensuring reliable data integrity under concurrent access.
- Owned the project end-to-end as sole developer , requirements gathering, iteration, deployment, and client handoff - for a live production client.

Infosys Springboard | AI & Data Science Intern — ***Oct 2025– Dec 2025***

- Developed a full-stack student analytics platform using FastAPI and Next.js, enabling tracking of 1,000+ study sessions and delivering personalised performance insights.
- Created 12+ REST APIs for managing study sessions, student data, OTP-based authentication, and admin analytics workflows.
- Built a machine learning pipeline using K-Means clustering (K=3) on 5+ behavioural features to segment students into performance categories, improving identification of at-risk students.
- Enabled data processing and visualisation pipelines using Pandas, Scikit-learn, and charting tools, supporting real-time dashboards and reducing analysis time by ~30%.

Edunet Foundation (AICTE & IBM SkillsBuild) | *AI & Cloud Intern* — ***July 2025– Aug 2025***

- Assembled an LLM-powered conversational AI agent using IBM watsonx.ai, implementing LangGraph-based orchestration with ReAct architecture for reasoning and tool invocation.
- Integrated Granite-3-3-8B-Instruct LLM with optimised decoding parameters, enabling context-aware itinerary generation across 500+ multi-turn interactions.
- Architected a stateful multi-turn conversation system with orchestrated decision-making, improving response relevance and task completion across complex user queries.
- Engineered modular Python-based tool calling and workflow orchestration, enabling scalable and extensible travel planning with structured outputs.

Xyronix Labs Pvt Ltd | *SDE Intern* — ***Jan 2025– June 2025***

- Built and maintained RESTful APIs using Django, handling authentication, data retrieval, and third party service integration.
- Collaborated on cloud-based service integration with OAuth2 authentication, enabling secure third party login for end users.
- Partnered with frontend and DevOps teams to deploy APIs into live systems via CI/CD pipelines, validating integration and resolving blockers end-to-end.

KEY PROJECTS

Emotion-Aware Wellness Chatbot | Tech stack: **Python, Hugging Face Transformers, Groq LLM** — **February 2026**

- Established an NLP-driven chatbot using transformer-based models (BERT/RoBERTa) to classify user sentiment and emotions across 5+ categories with real-time inference.
- Applied Groq LLM for context-aware response generation, reducing irrelevant responses and improving personalisation quality across 1,000+ interactions.
- Designed a modular inference pipeline (emotion detection → context mapping → response generation), reducing end-to-end response latency to <300 ms.
- Implemented session-based context handling and prompt engineering techniques to maintain conversational continuity and improve response coherence.
- Structured an end-to-end system delivering personalised coping strategies and mood tracking, increasing user engagement and retention in simulated test scenarios.

Exoplanet Data Analysis & Visualisation | *Tech Stack:* ***Python, Pandas, Matplotlib, Seaborn*** — ***October 2025***

- Performed data preprocessing and feature engineering on NASA exoplanet datasets (10,000+ records), handling missing values, normalisation, and outlier detection.
- Conducted exploratory data analysis using statistical methods and correlation mapping to identify relationships between planetary features (mass, radius, orbital parameters).
- Programmed advanced visualisation pipelines (multi-dimensional plots, distribution analysis, and habitability zone mapping) to extract and present key insights.

CERTIFICATIONS

- **Cloud Application Development –** *IBM* (IBM Cloud, REST APIs, Microservices Architecture, Cloud Deployment)
- **Machine Learning using R –** *IBM* (Supervised Learning, Regression & Classification, Statistical Modeling, Model Evaluation)